

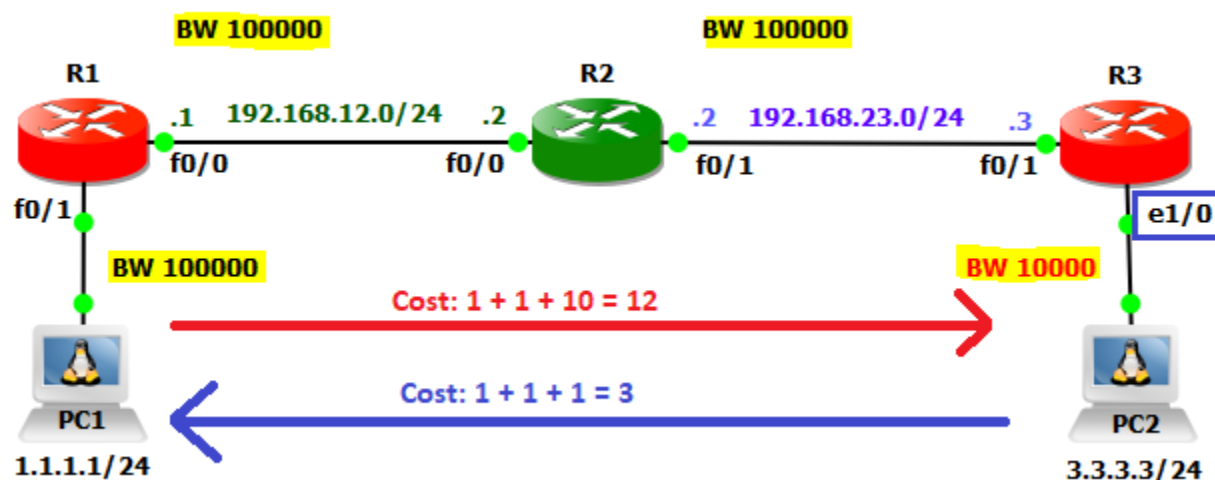
OSPF Metric:

OSPF uses a metric called **cost**, which is based on the bandwidth of an interface.

Cost = Reference Bandwidth / Interface Bandwidth

the reference bandwidth is a default value on Cisco routers which is a **100Mbit (10⁸)** interface. Divide the reference bandwidth by the bandwidth of the interface and get the cost. The lower the cost the better the path is. In route, that has lowest cumulative cost value between source and destination will be selected for routing table. If two paths are equal cost, OSPF will use both paths and will load balance among them 50/50.

Default Cost of Interfaces			
Interface Type	Bandwidth	Metric Calculation	Cost
Ethernet Link	10Mbps	$100000000/10000000 = 10$	10
FastEthernet Link	100Mbps	$100000000/100000000 = 1$	1
Serial Link	1544Kbps	$100000000/1544000 = 64.76$	64
Gigabyte Link	1 Gbps		1
10 Gigabit Link	10 Gbps		1
40 Gigabit Link	40 Gbps		1
100 Gigabit Link	100 Gbps		1



From PC1 to PC2 total OSPF cost is **12** because in the way one Ethernet link which is connected between R3 to PC2. **Cost is counting from outgoing interfaces to reach destination.** Therefore, from PC1 to PC2 OSPF total cost is cost is 12 of R1, R2 and R3 outgoing interfaces.

```
R1#show ip route ospf
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

Gateway of last resort is not set

3.0.0.0/24 is subnetted, 1 subnets

O 3.3.3.0 [110/12] via 192.168.12.2, 00:12:49, FastEthernet0/0

O 192.168.23.0/24 [110/2] via 192.168.12.2, 00:12:49, FastEthernet0/0

```
R1#show ip ospf int f0/0 | sec Cost
```

Process ID	1	Router ID	192.168.12.1	Network Type	BROADCAST	Cost	1
Topology-MTID	0	Cost	1	Disabled	no	Shutdown	no
Topology Name	Base						

```
R2#show ip ospf interface f0/1 | sec Cost
```

Process ID	1	Router ID	192.168.23.2	Network Type	BROADCAST	Cost	1
Topology-MTID	0	Cost	1	Disabled	no	Shutdown	no
Topology Name	Base						

```
R3#show ip ospf interface e1/0 | sec Cost
```

Process ID	1	Router ID	192.168.23.3	Network Type	BROADCAST	Cost	10
Topology-MTID	0	Cost	10	Disabled	no	Shutdown	no
Topology Name	Base						

From PC2 to PC1 total OSPF cost is 3 as all link are FastEthernet. Cost is counting from outgoing interfaces to reach destination. Therefore, from PC2 to PC1 OSPF total cost is cost is 3 of R3, R2 and R1 outgoing interfaces.

```
R3#show ip route ospf
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

1.0.0.0/24 is subnetted, 1 subnets

O 1.1.1.0 [110/3] via 192.168.23.2, 00:25:43, FastEthernet0/1

O 192.168.12.0/24 [110/2] via 192.168.23.2, 00:25:43, FastEthernet0/1

```
R3#show ip ospf interface f0/1 | sec Cost
```

Process ID	1	Router ID	192.168.23.3	Network Type	BROADCAST	Cost	1
Topology-MTID	0	Cost	1	Disabled	no	Shutdown	no
Topology Name	Base						

```
R2#show ip ospf interface f0/0 | sec Cost
```

Process ID	1	Router ID	192.168.23.2	Network Type	BROADCAST	Cost	1
Topology-MTID	0	Cost	1	Disabled	no	Shutdown	no
Topology Name	Base						

```
R1#show ip ospf int f0/1 | sec Cost
```

Process ID	1	Router ID	192.168.12.1	Network Type	BROADCAST	Cost	1
Topology-MTID	0	Cost	1	Disabled	no	Shutdown	no
Topology Name	Base						